MONTANA

DATA COMPANY

CALCULATING YOUR TRUE RTO & RPO

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Introduction

A business-led framework for defining recovery time and recovery point objectives.

Many organizations define Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) based on technical assumptions instead of business impact.

This often results in recovery targets that are either too expensive to maintain or dangerously inadequate during a disruption.

This guide provides a business-led framework for determining realistic and defensible RTO and RPO targets aligned to operational risk, customer impact, legal obligations, and financial exposure.

1. Understanding RTO and RPO

RTO (Recovery Time Objective): The maximum acceptable amount of time a system, process, or service can remain unavailable after an incident.
RPO (Recovery Point Objective): The maximum acceptable amount of data loss measured in time before the disruption occurred.
RTO answers: 'How long can we be down?'
RPO answers: 'How much data can we afford to lose?'

2. Why Technical Targets Fail

Many IT teams define recovery objectives based on infrastructure capabilities rather than business consequences.

Common failures include:

- Assigning identical RTO/RPO values to all systems
- Using vendor defaults instead of operational realities
- Ignoring manual workarounds
- Not validating assumptions with business units
- Failing to account for regulatory or contractual obligations

3. Start With Business Impact

Before setting targets, identify the operational consequences of downtime and data loss.

Assess impacts across:

- Revenue loss
- Customer service disruption
- Regulatory exposure
- Reputational damage
- Operational paralysis
- Health and safety considerations
- Supply chain disruption

4. Identify Critical Business Processes

Document the business processes that generate revenue, support customers, ensure compliance, or enable operations.

Examples include:

- Customer billing
- Payroll processing
- Manufacturing operations
- CRM access
- Email and collaboration systems
- Financial transaction processing

5. Map Systems to Business Functions

Many organizations protect systems without understanding which business functions depend on them.

Create dependency maps linking:

- Business process
- Supporting applications
- Infrastructure dependencies
- Third-party services
- Data repositories
- Key personnel

6. Define Downtime Tolerance

Interview business stakeholders to determine acceptable downtime thresholds.

Useful questions include:

- What happens after 15 minutes of outage?
- What happens after 1 hour?
- When does customer impact become severe?
- At what point does revenue loss become unacceptable?
- When do legal or SLA breaches occur?

7. Define Data Loss Tolerance

Determine how much lost data the business can realistically tolerate.

Consider:

- Financial transactions
- Customer orders
- Audit records
- Compliance data
- Operational records
- Productivity impacts from rework

8. RTO and RPO Classification Matrix

Example classification model:

Tier 1 – Mission Critical
RTO: Minutes to 1 hour
RPO: Near zero to 15 minutes

Tier 2 – Business Critical
RTO: 4–8 hours
RPO: 1–4 hours

Tier 3 – Important
RTO: 24 hours
RPO: 8–24 hours

Tier 4 – Non-Critical
RTO: Multiple days
RPO: 24+ hours

9. Cost vs Recovery Expectations
Lower RTO and RPO targets significantly increase infrastructure and operational costs.

Business leaders must understand the trade-offs between:
- Replication technologies
- Backup frequency
- High availability architectures
- Disaster recovery sites
- Cloud failover solutions
- Operational complexity

10. Validate Against Real Scenarios
Test whether your targets are realistic under actual failure conditions.

Scenarios should include:
- Ransomware attacks
- Data corruption
- Cloud outages
- Human error
- Power failure
- Internet disruption
- Third-party supplier failure

11. Recovery Dependency Risks
Recovery targets often fail because dependencies are overlooked.

Validate dependencies including:
- Identity systems
- DNS services
- Network connectivity
- MFA platforms
- Backup infrastructure
- Vendor support availability

12. Governance and Accountability
RTO and RPO ownership should not belong solely to IT.

Assign accountability across:
- Executive leadership
- Business unit owners
- IT operations
- Risk management

- Compliance teams

13. Recovery Objective Review Cycle
Recovery objectives should be reviewed regularly.

Review triggers include:
- New systems or applications
- Business expansion
- Regulatory changes
- Mergers and acquisitions
- Major infrastructure changes
- Security incidents

14. Common Mistakes
Avoid these common pitfalls:
- Setting unrealistic recovery expectations
- Failing to test recovery procedures
- Ignoring cloud provider limitations
- Treating backups as disaster recovery
- Assuming replication equals recoverability
- Not involving business stakeholders

15. Practical Workshop Template
For each critical service, document:
- Business owner
- Operational impact of downtime
- Maximum tolerable outage
- Maximum tolerable data loss
- Current recovery capability
- Required recovery capability
- Identified gaps
- Responsible stakeholders

16. Executive Summary Checklist
Use this checklist to validate your recovery strategy:
- ☐ Business processes prioritized
- ☐ Recovery dependencies identified
- ☐ Business-approved RTO values defined
- ☐ Business-approved RPO values defined
- ☐ Recovery assumptions documented
- ☐ Disaster recovery testing performed
- ☐ Executive ownership assigned
- ☐ Recovery gaps identified and funded

Conclusion
Effective recovery planning is fundamentally a business decision supported by technology.

Organizations that align RTO and RPO targets to real operational impact are better positioned to withstand ransomware attacks, infrastructure failures, and operational disruptions.

Recovery objectives should be measurable, tested, approved by business leadership, and reviewed as the organization evolves.

Contact Us

If you would like to discuss your organisation's specific requirements, risks, compliance obligations, cybersecurity posture, business continuity strategy, or operational resilience objectives in greater detail, we invite you to contact Montana Data Company.

Our team can assist with tailored assessments, strategic guidance, governance support, and practical implementation aligned to your organisation's unique operational and regulatory environment.

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